# Md. Abdur Rahman

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## Education

**Military Institute of Science and Technology** — *MSc in Computer Science and Engineering (ongoing)* *Apr 2026 – Present*

**Southeast University** — *BSc in Computer Science and Engineering* *Oct 2021 – Oct 2025*

- CGPA: 3.87/4.00 — **Last 60 Credits CGPA: 3.97/4.00**
- **Relevant Coursework:** Artificial Intelligence, Image Processing, Data Mining, Research Methodology

## Experience

**Research Assistant** — *Islamic University of Technology (IUT), OIC* *Gazipur, BD*
*Nov 2025 – Present*

- Engineered a Federated Learning framework for fintech data with a novel aggregation technique for non-IID settings.
- Pioneered hyperbolic Graph Neural Networks for blockchain fraud detection and causal inference methods for high-dimensional auditing.
- Authored five manuscripts accepted in Q1 journals on decentralized AI, graph-based fraud detection.

**Research Assistant** — *CAIR, Southeast University* *Dhaka, BD*
*Jan 2025 – Oct 2025*

- Designed hybrid transformer-based architectures for medical image segmentation and classification, fusing convolutional and attention mechanisms to improve diagnostic accuracy on MRI and X-ray datasets.
- Integrated explainable AI methods (SHAP, LIME) into clinical decision-support pipelines, enabling interpretable model outputs for high-stakes diagnostic settings.
- Authored five papers: three accepted at peer-reviewed conferences, two under review in Q1 journals on medical AI and interpretable deep learning.

## Selected Publications

***J*** *= Journal* ***C*** *= Conference* ***W*** *= Workshop*

[J.1] **MangoLeafNet-XAI: An Attention-Enhanced Deep Learning Architecture for Accurate and Interpretable Mango Leaf Disease Classification** [Paper]
***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Farzan Majeed Noori, Md Zia Uddin, Abdul Kadar Muhammad Masum
Frontiers in Plant Science [Q1]

[J.2] **Integrating Domain-Specific Convolutional Features with Swin Transformers for Transparent Cervical Cancer Diagnostics** [Paper]
***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Md Fokrul Islam KHAN, Md Zia Uddin, Ulysse Teller Masao Côté-Allard, Abdul Kadar Muhammad Masum
Frontiers in Oncology [Q2]

[J.3] **FedMal-XAI: An Explainable Federated Vision Transformer Leveraging Knowledge Distillation for Privacy-Preserving Malaria Detection** [Paper]
Md. Tofael Ahmed Bhuiyan, ***Md. Abdur Rahman***, Md Fokrul Islam Khan, Farzan M. Noori, Abdul Kadar Muhammad Masum
Frontiers in Public Health [Q1]

[J.4] **DFU-GCNet: A Global Context-Enhanced Inception Network for Robust and Interpretable Diabetic Foot Ulcer Classification** [Paper]
Md. Tofael Ahmed Bhuiyan, ***Md. Abdur Rahman***, Farzan M. Noori, Abdul Kadar Muhammad Masum
Frontiers in Digital Health [Q1]

[J.5] **Privacy-Preserving ESG Risk Prediction via Federated Hybrid Neural-Boosting Ensembles** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Ramona Birău, Virgil Popescu
IEEE Accessc [Q1]

[J.6] **Predicting Corporate Bankruptcy Through a Scalable and Interpretable Knowledge Distillation Framework** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Thanh Lam Nguyen, Hasan Dinçer, Serhat Yuksel
Discover Artificial Intelligence [Q1]

[J.7] **EcoStack-Pro: An Adaptive Federated Learning Framework for Interpretable ESG Auditing Across Heterogeneous Industrial Sectors** [Paper]
Md. Abul Kalam Azad, Abdul Kadar Muhammad Masum, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Farzan M. Noori, Md Zia Uddin
Frontiers in Artificial Intelligence [Q1]

[J.8] **Bridging Data Silos in Corporate Governance: A Hierarchical Stacking Ensemble with Federated Dynamic Aggregation** [Paper]
Abdul Kadar Muhammad Masum, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**
Applied Computational Intelligence and Soft Computing [Q1]

[J.9] **Beyond Short-Term Noise: Robust Long-Horizon Bankruptcy Prediction Using Gated Dual-Branch Transformers** [Paper]
Rezwanul Huque Khan, Md. Abul Kalam Azad, Md. Tofael Ahmed Bhuiyan, **Md. Abdur Rahman**, Faisal Md. Abdur Rahman, Abdul Kadar Muhammad Masum
Applied Computational Intelligence and Soft Computing [Q1]

[C.1] **FAKD-XAI: Feature-Aligned Knowledge Distillation with Explainable AI for Efficient Brain Tumor Classification** [Paper]
***Md. Abdur Rahman***, Sabik Aftahee, Md Ashiqur Rahman, Lamim Zakir Pronay
International Conference on Data Science, AI and Applications (ICDSAIA 2025)

[C.2] **X-ArsenicDistill: An Efficient and Interpretable Framework for Arsenicosis Classification via Knowledge Distillation** [Paper]
***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Mohammad Mahmudur Rahman, Iffat Sania Hossain, Abu Kowshir Bitto, Prof. Dr. Abdul Kadar Muhammad Masum
28th International Conference on Computer and Information Technology (ICCIT 2025)

[C.3] **DeepSea-Net: A YOLO Based Framework for Real-Time Detection and Classification of Underwater Plastic Pollution** [Paper]
Progga Parmita Roy, Sabbir Sarder Munna, Md. Sazed Ul Karim, ***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Md. Joseb Siddique Siam
7th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2025)

[W.1] **Do LLMs Distinguish Between Halal and Haram? Benchmarking Islamic Cultural Alignment in General vs. Arabic-Centric SLMs** [Paper]
***Md Abdur Rahman***, Md Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
6th Muslims in ML Workshop, ICML 2026

[W.2] **Tokenizer Transplantation: Mitigating Autoregressive Collapse in Edge-Efficient Bengali ASR** [Paper]
Sanjid Hasan, ***Md Abdur Rahman***
6th Muslims in ML Workshop, ICML 2026

[W.3] **SFL-MTKD++: Split ViT Federated Learning with Adaptive Multi-Teacher Distillation for Interpretable Malaria Diagnosis** [Paper]
Md Tofael Ahmed Bhuiyan, ***Md Abdur Rahman***, Abdul Kadar Muhammad Masum
6th Muslims in ML Workshop, ICML 2026

[W.4] **AlzFed-XAI: High-Fidelity Interpretable Alzheimer's Diagnosis with Privacy-Preserving Federated Learning** [Paper]
***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
5th Muslims in ML Workshop, NeurIPS 2025

[W.5] **ChakmaBridge: A Five-Way Parallel Corpus for Navigating the Script Divide in an Endangered Language** [Paper]
***Md. Abdur Rahman***, Md. Tofael Ahmed Bhuiyan, Abdul Kadar Muhammad Masum
2nd Bangla Language Processing (BLP) Workshop, AACL-IJCNLP 2025

[W.6] **BhasaBodh: Bridging Bangla Dialects and Romanized Forms through Machine Translation** [Paper]
Md. Tofael Ahmed Bhuiyan, ***Md. Abdur Rahman***, Abdul Kadar Muhammad Masum
2nd Bangla Language Processing (BLP) Workshop, AACL-IJCNLP 2025

[W.7] **Detecting Manipulation in Ukrainian Telegram: A Transformer-Based Approach to Technique Classification and Span Identification** [Paper]
***Md. Abdur Rahman***, Md Ashiqur Rahman
4th Ukrainian Natural Language Processing Workshop (UNLP), ACL 2025

## Under Review Publications

1. **HA-DynAU-Net: A Hybrid Attention Dynamic Aware U-Net for Precise Skin Lesion Segmentation** [1st Author]
   Submitted to *IEEE Open Journal of the Computer Society*
2. **BEACON: A Novel Cross-Modal Architecture for High-Precision and Explainable Brain Tumor Classification Using Medical-Specific Attention** [1st Author]
   Submitted to *PLOS ONE*
3. **HybridTransUNet: Fusing Vision Transformers with Attentive U-Net for Accurate and Interpretable Brain Tumor Segmentation** [1st Author]
   Submitted to *Engineering Reports*
4. **Rerank-RRG: Cross-Modal Phrase Reranking for Retrieval-Augmented Radiology Report Generation** [2nd Author]
   Submitted to *29th International Conference on Computer and Information Technology (ICCIT)*

## Experience with Research Grant

*Assisted in proposal preparation for the following research grants:*

- **Closing the Employability Gap: A Machine Learning–Driven Framework to Predict and Improve Student Placement Outcomes at University** *August 2026*
  Co-PI, IRT Research Grant, Southeast University
  Funded amount: $4,000
- **Bridging the Gap: A Predictive Analytics Framework for Enhancing Engineering Student Employability and Placement Outcomes in Bangladesh** *September 2026*
  Team Member, Bangladesh Accreditation Council (BAC) Research Grant
  Funded amount: $3,000

## Honors & Awards

- **3rd Position (Poster Presentation)**, Student Research Day 2025 [Certificate]
- **Champion**, Fragments of Feeling: A Contextual Emotion Challenge (2025) [GitHub]
- **1st Runners-up**, CalorieQuest: Predicting Energy Burn with Data (2025) [GitHub]
- **2nd Runners-up**, DU AI Challenge (2025) [GitHub]
- **2nd Runners-up**, REACT 2026 Datathon, hosted by IEEE SEU Student Branch [GitHub]
- **Finalist (Top 10)**, CUET ETE Datathon 2025 [GitHub]

## Academic Service

- **Program Committee**, 21st BEA Workshop co-located with **ACL 2026**
- **Reviewer**, 7th Muslims in ML Workshop at **NeurIPS 2026**
- **Reviewer**, Frontiers in Artificial Intelligence
- **Reviewer**, Frontiers in Medicine
- **Reviewer**, Green Technologies and Sustainability

## Extracurricular Activities

**Research Executive (ExCom)**, IEEE Computer Society Bangladesh Chapter Graduate Student Member Forum *2026*

**Sub-Executive (Research & Development)**, IEEE Computer Society Southeast University Student Branch Chapter *2025*

## Skills

**Languages:** C, C++, Java, Python
**Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, PyTorch, HuggingFace, OpenCV, Django, FastAPI, Spring Boot
**Technologies:** Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Git/GitHub, Cloud Computing, LaTeX

## Professional Certification

- **AWS Certified Cloud Practitioner**, Amazon Web Services *Dec 2025 – Dec 2028*
- **GitHub Foundations Certification**, GitHub *Nov 2024 – Nov 2027*